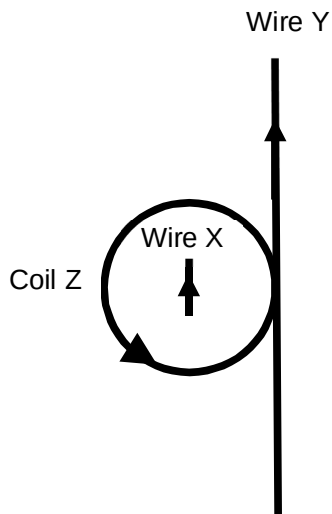


|           |                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------|
| <b>24</b> | The diagram below shows a setup of two current-carrying wires X and Y, and a single-turn coil Z. |
|-----------|--------------------------------------------------------------------------------------------------|



Wire X has length 0.050 m with a current of 1.0 A flowing through it. Wire Y has length 2.0 m with a current of 0.50 A. Coil Z has a diameter of 0.20 m with a current of 1.5 A.

What is the resultant force on wire X if the centre of coil Z is on wire X?

|          |                                |
|----------|--------------------------------|
| <b>A</b> | $5.0 \times 10^{-8} \text{ N}$ |
| <b>B</b> | $4.2 \times 10^{-7} \text{ N}$ |
| <b>C</b> | $5.2 \times 10^{-7} \text{ N}$ |
| <b>D</b> | $9.4 \times 10^{-6} \text{ N}$ |

**Answer: C**

By right-hand grip rule, the magnetic flux densities due to wire Y and due to coil Z are **both** out of the page.

Hence, the resultant magnetic flux density experienced by wire X

$$= \frac{\mu_0 NI}{2r} + \frac{\mu_0 I}{2\pi d} = \frac{4\pi \times 10^{-7} \times 1 \times 1.5}{2(0.1)} + \frac{4\pi \times 10^{-7} \times 0.5}{2\pi(0.1)}$$

$$= 1.042 \times 10^{-5} \text{ T}$$

$$\text{Therefore, the resultant force on wire X} = BIL = 1.042 \times 10^{-5} \times 1.0 \times 0.05 = 5.2 \times 10^{-7} \text{ N}$$